Q1a) Explain Wumpus World Environment with PEAS description.

Wumpus World Environment

Wumpus World is a classic Artificial Intelligence problem used to represent a knowledge-based agent environment. It consists of: Rooms connected in a grid, A dangerous creature called Wumpus, Pits, Gold.

The goal of the agent is: Find the gold, Avoid pits and Wumpus, Safely return back.

Features of Wumpus World

1. Environment is represented as grid (usually 4×4).
2. Agent can move in four directions.
3. Wumpus can kill the agent.
4. Pits cause agent to fall.
5. Gold gives reward.

Indicators in Wumpus World

Indicator	**Meaning**
Breeze	Pit nearby
Stench	Wumpus nearby
Glitter	Gold present
Bump	Hit wall
Scream	Wumpus killed

Wumpus World Diagram

```
-----------------
|   | P |   | G |
|---|---|---|---|
|   | W |   | P |
|---|---|---|---|
|   |   | P |   |
|---|---|---|---|
| A |   |   |   |
-----------------
```

Where: A → Agent, W → Wumpus, P → Pit, G → Gold

PEAS Description

PEAS stands for: Performance Measure, Environment, Actuators, Sensors

1. Performance Measure - Measures success of agent.

Goals: Find gold, Avoid death, Minimize moves

Rewards: Gold found → Positive reward

Death → Negative reward

2. Environment - The environment includes: Rooms, Wumpus, Pits, Gold, Walls

Characteristics: Partially observable, Sequential, Static, Discrete

3. Actuators - Actions performed by agent: Move Forward, Turn Left, Turn Right, Grab, Shoot, Climb

4. Sensors - Agent senses: Breeze, Stench, Glitter, Bump, Scream

Applications of Wumpus World - Knowledge-based systems, Robotics navigation, AI reasoning, Intelligent agents

Advantages - Demonstrates logical reasoning, Useful for AI learning, Models uncertain environments

Q2b) What is knowledge representation in propositional logic? Compare and contrast PL and FOL.

Knowledge Representation (KR) is a method of storing knowledge in a form that AI systems can understand and use for reasoning.

It helps machines: Represent facts, Make decisions, Perform inference

Propositional Logic (PL) is a formal system in which statements are represented as propositions that are either: True or False.

Representation in PL -

Example statements: P:"It is raining" Q:"Road is wet"

Logical relation: $P \rightarrow Q$

Meaning: "If it is raining, then road is wet."

Components of PL

1. Propositions
2. Logical operators: AND (∧), OR (∨) ,NOT (¬), Implication (→).

Advantages of PL - Simple representation, Easy reasoning, Used in rule-based systems

Limitations of PL - Cannot represent relationships, Cannot handle variables and quantifiers, Less expressive

First Order Logic (FOL) extends PL by adding: Predicates, Variables, Quantifiers

Example: $Human(x) \rightarrow Mortal(x)$
Meaning: "All humans are mortal."

Quantifiers in FOL -

Universal Quantifier $\forall x$ Means "for all".

Existential Quantifier $\exists x$ Means "there exists".

Comparison between PL and FOL

Feature	**Propositional Logic (PL)**	**First Order Logic (FOL)**
Basic Unit	Proposition	Predicate
Expressiveness	Less	More
Variables	Not supported	Supported
Quantifiers	Not available	Available
Relationship Representation	Not possible	Possible
Complexity	Simple	More complex
Example	$P \rightarrow Q$	$Human(x) \rightarrow Mortal(x)$
Reasoning Power	Limited	Powerful
Applications	Digital circuits, Simple expert systems, Rule-based AI	NLP, Semantic web, Knowledge-based systems, Intelligent reasoning

Propositional Logic is simple but limited. First Order Logic is more expressive and powerful. FOL is preferred for complex AI knowledge representation systems.

a) Explain different inference rules in First Order Logic (FOL) with suitable example.

First Order Logic (FOL) is an extension of Propositional Logic that uses: Predicates, Variables, Quantifiers. It is used for knowledge representation and reasoning in AI.

Inference Rules are logical rules used to derive new facts from existing facts.

1. Modus Ponens

Rule -

If: $P \rightarrow Q$ and P is true

then: Q is true.

Example - $Rain \rightarrow WetRoad \qquad Rain$

Therefore: $WetRoad$

2. Universal Instantiation (UI)

Allows replacing universally quantified variable with a specific constant.

Rule –

From: $\forall x\, Human(x)$

infer: $Human(John)$

Example - "All humans are mortal."

$$\forall x(Human(x) \rightarrow Mortal(x))$$

Then: $Human(Ram) \rightarrow Mortal(Ram)$

3. Existential Instantiation (EI)

Replaces existential variable with a new constant.

Rule -

From: $\exists x\, Student(x)$

infer: $Student(A)$

where A is a new object.

Example - "There exists a student who passed."

$$\exists x\, Passed(x)$$

Infer: $Passed(Rahul)$

4. Universal Generalization (UG)

If statement is true for arbitrary object, it is generalized for all objects.

Example -

If: $Mortal(Ram)$
then infer: $\forall x\, Mortal(x)$ (under proper conditions)

5. Existential Generalization (EG)

If property holds for one object, infer that there exists such an object.

Example - $Student(Ram)$
Infer: $\exists x\, Student(x)$

6. Resolution Rule

Used in theorem proving and automated reasoning.

Combines clauses to derive contradiction or conclusion.

Example -

Given: $P \vee Q$

$\neg Q$

Infer: P

Applications of FOL Inference - Expert systems, Natural Language Processing, Robotics, Automated theorem proving, Knowledge-based systems

Q4b) What is an agent? Explain knowledge-based agent with architecture diagram. Also state significance of inference engine.

An agent is an entity that: Perceives environment using sensors, Acts upon environment using actuators.

It performs actions intelligently to achieve goals.

"An agent perceives its environment through sensors and acts upon that environment through actuators."

Examples of Agents – Robot, Chatbot, Self-driving car, ATM machine, Virtual assistant, Human

Components of Agent

1. Sensors
2. Environment
3. Actuators

A Knowledge-Based Agent is an intelligent agent that uses stored knowledge and reasoning to make decisions.

It: Stores facts in knowledge base, Uses inference engine to derive conclusions, Updates knowledge continuously

Components of Knowledge-Based Agent

1. Knowledge Base (KB)

Stores Facts, Rules, Information about environment

Example: $Human(Ram)$

$Human(x) \rightarrow Mortal(x)$

2. Inference Engine -

Reasoning mechanism that: Applies inference rules,

Derives new knowledge, Makes decisions

3. Sensors - Collect information from environment.

Examples: Camera, Keyboard, Microphone

4. Actuators - Perform actions.

Examples: Robot arms, Display, Motors

```
 Environment
      ↑
   Sensors
      ↑
 ----------------
| Knowledge    |
|    Base      |
 ----------------
      ↑
 ----------------
| Inference    |
|   Engine     |
 ----------------
      ↓
   Actuators
      ↓
  Environment
```

Working of Knowledge-Based Agent

1. Agent perceives environment.
2. Information stored in KB.
3. Inference engine applies reasoning.
4. Agent decides action.
5. Action executed using actuators.

Significance of Inference Engine

Inference engine is the "brain" of knowledge-based system.

Functions of Inference Engine - Applies logical rules, Derives conclusions, Solves queries, Supports decision making, Performs reasoning

Importance - Converts knowledge into intelligent action, Automates reasoning, Improves decision-making capability, Enables expert system functionality

Types of Inference -

1. Forward Chaining - Data-driven, Starts from facts

2. Backward Chaining - Goal-driven, Starts from goal

Applications of Knowledge-Based Agents - Medical diagnosis systems, Expert systems, Robotics, AI assistants, Decision support systems

Advantages - Intelligent reasoning, Handles complex problems, Reusable knowledge, Better decision-making.

Q3a) Define Knowledge Base and Sentence. Describe Wumpus World Environment along with brief description to find out the agent. Explain Task Environment.

A Knowledge Base (KB) is a collection of facts, rules, and information stored in a system which is used by an intelligent agent for reasoning and decision making.

It contains: Facts about environment, Rules and relationships, Logical statements

Example -

Human(Ram)
Human(x) → Mortal(x)

From above knowledge, system can infer: Mortal(Ram)

Sentence in AI - A sentence is a statement written in a formal knowledge representation language that expresses some fact or rule.

Sentences can be: True, False

Examples - Propositional Logic $P \rightarrow Q$

"If it rains, roads become wet."

First Order Logic $Human(Ram)$

"Ram is a human."

Wumpus World Environment - ...

How Agent Finds Goal - The agent uses percepts from sensors.

Example -

1. **Breeze Detected -** Pit exists in nearby square.

2. **Stench Detected -** Wumpus nearby.

3. **Glitter Detected -** Gold found.

Agent uses logical reasoning to: Avoid dangerous cells, Move toward safe cells, Reach gold safely

Knowledge-Based Reasoning in Wumpus World

Example Rule:

$$Breeze(x, y) \rightarrow PitNearby(x, y)$$
$$Stench(x, y) \rightarrow WumpusNearby(x, y)$$

The agent stores such rules in Knowledge Base and uses inference engine for reasoning.

Task Environment

Task Environment specifies the working environment of an intelligent agent. It is described using PEAS. ...

Q3b) Represent the following into First Order Logic (FOL).

i) All employees earning Rs.45000 or more pay tax.

$$\forall x[(Employee(x) \land Salary(x) \geq 45000) \rightarrow PayTax(x)]$$

ii) Sita is a marine engineer and she is also an artist.

$$MarineEngineer(Sita) \land Artist(Sita)$$

iii) Children love ice cream.

$$\forall x[Child(x) \rightarrow Loves(x, IceCream)]$$

iv) If humidity is high and temperature is high then a person cannot feel comfortable.

$$\forall x[(HighHumidity \land HighTemperature) \rightarrow \neg Comfortable(x)]$$

v) Puppies are cute.

$$\forall x[Puppy(x) \rightarrow Cute(x)]$$

vi) If AB and AC are equal, then angle B and angle C are equal.

$$Equal(AB, AC) \rightarrow Equal(AngleB, AngleC)$$

vii) ABC is an equilateral triangle.

An equilateral triangle has all sides equal.

$$EquilateralTriangle(ABC)$$

OR more specifically:

$$Equal(AB, BC) \land Equal(BC, CA)$$

Advantages of FOL Representation - Expressive representation, Supports reasoning, Represents objects and relations, Used in AI knowledge systems

Q4a) Write short notes on the following:

i) Syntax and Semantics

Syntax defines the rules for writing valid statements or expressions in a logical language.

It specifies: Symbols used, Structure of statements, Formation rules

Example in Propositional Logic $P \rightarrow Q$ is syntactically correct.

But: $\rightarrow PQ$ is syntactically incorrect.

Components of Syntax -

1. Symbols

2. Variables
3. Logical operators
4. Formation rules

Semantics defines the meaning or interpretation of logical statements.

It determines whether a statement is: True or False

Example P:"It is raining" Q:"Road is wet"

Statement: $P \to Q$

Meaning: "If it rains, road becomes wet."

Applications - Knowledge representation, Programming languages, AI reasoning systems

ii) Propositional Logic vs First Order Logic

Feature	**Propositional Logic**	**First Order Logic**
Basic Unit	Proposition	Predicate
Variables	Not supported	Supported
Quantifiers	Not available	Available
Expressiveness	Less	More
Relationship Representation	Not possible	Possible
Complexity	Simple	Complex
Example	$P \to Q$	$Human(x) \to Mortal(x)$

iii) Knowledge Engineering Process in First Order Logic

Knowledge Engineering is the process of building knowledge-based systems using logical representation and reasoning.

Steps in Knowledge Engineering Process -

1. Identify the Task - Determine problem domain.

Example: Medical diagnosis, Robotics

2. Gather Knowledge - Collect: Facts, Rules, Expert knowledge

3. Choose Vocabulary - Define: Predicates, Functions, Constants

Example: $Human(x), Mortal(x)$

4. Encode General Knowledge - Represent knowledge in FOL.

Example: $\forall x(Human(x) \rightarrow Mortal(x))$

5. Encode Specific Facts

Example: $Human(Ram)$

6. Query the System - Ask questions to inference engine.

Example: $Mortal(Ram)?$

7. Debug and Improve - Check correctness and improve knowledge base.

Applications - Expert systems, NLP, Robotics, Intelligent agents

Q4b) Show the following sentences are valid or not.

i) $(P \land Q) \rightarrow (P \lor Q)$

We check truth table.

P	**Q**	$P \land Q$	$P \lor Q$	**Result**
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

Since all results are True, $(P \land Q) \rightarrow (P \lor Q)$ is **VALID**.

ii) $(\neg A \lor B) \land (\neg B \lor C) \rightarrow (\neg A \lor C)$

This represents transitive implication.

Simplification

$(\neg A \lor B) = A \rightarrow B$

$(\neg B \lor C) = B \rightarrow C$

Thus: $(A \rightarrow B) \land (B \rightarrow C) \rightarrow (A \rightarrow C)$ This is logically true.

Hence, $(\neg A \lor B) \land (\neg B \lor C) \rightarrow (\neg A \lor C)$ is **VALID**.

Q3b) Represent the following facts in Predicate Logic.

Predicate Logic Representation -

1. If a triangle is equilateral then it is isosceles.

$$\forall x[Equilateral(x) \rightarrow Isosceles(x)]$$

2. If a triangle is isosceles, then its two sides AB and AC are equal.

$$\forall x[Isosceles(x) \rightarrow Equal(AB, AC)]$$

3. If AB and AC are equal, then angle B and angle C are equal.

$$Equal(AB, AC) \rightarrow Equal(AngleB, AngleC)$$

4. ABC is an equilateral triangle.

$$Equilateral(ABC)$$

Conclusion from Above Facts

Using inference: $Equilateral(ABC) \rightarrow Isosceles(ABC)$

Therefore: $Isosceles(ABC)$

Then: $Equal(AB, AC)$

Finally: $Equal(AngleB, AngleC)$

Thus proved using predicate logic.

Q4a) Write the following sentences in First Order Logic (FOL)

i) All birds fly - This statement uses **Universal Quantifier** (∀)

$$\forall x[Bird(x) \rightarrow Fly(x)]$$

Meaning: "For every x, if x is a bird then x can fly."

ii) Some boys play cricket - This statement uses **Existential Quantifier** (∃)

$$\exists x[Boy(x) \land Plays(x, Cricket)]$$

Meaning: "There exists at least one boy who plays cricket."

iii) A first cousin is a child of a parent's sibling

$$\forall x \forall y[FirstCousin(x, y) \rightarrow ChildOf(x, SiblingOf(Parent(y)))]$$

OR simpler representation: $\forall x \forall y[(Sibling(Parent(x), Parent(y))) \rightarrow FirstCousin(x, y)]$

Meaning: "If parents of x and y are siblings, then x and y are first cousins."

iv) You can fool all the people some of the time, and some of the people all the time, but you cannot fool all the people all the time.

This sentence contains both:

- Universal quantifier (∀)
- Existential quantifier (∃)

Part 1: "You can fool all the people some of the time."

$$\forall x \exists t\, Fool(x, t)$$

Meaning: "For every person x, there exists some time t when x can be fooled."

Part 2: "You can fool some of the people all the time."

$$\exists x \forall t\, Fool(x, t)$$

Meaning: "There exists some people who can be fooled all the time."

Part 3: "You cannot fool all the people all the time."

$$\neg \forall x \forall t\, Fool(x, t)$$

Meaning: "It is not true that every person can be fooled at every time."

Final Combined Representation -

$$(\forall x \exists t\, Fool(x, t)) \land (\exists x \forall t\, Fool(x, t)) \land \neg(\forall x \forall t\, Fool(x, t))$$

Q4a) Write the following sentences in First Order Logic (FOL)

i) Every number is either negative or has a square root.

$$\forall x[Number(x) \rightarrow (Negative(x) \lor HasSquareRoot(x))]$$

Meaning: "For every x, if x is a number then it is either negative or has a square root."

ii) Every connected and circuit-free graph is a tree.

$$\forall x[(Connected(x) \land CircuitFree(x)) \rightarrow Tree(x)]$$

Meaning: "If a graph is connected and circuit-free, then it is a tree."

iii) Some people are either religious or pious.

$$\exists x[Person(x) \land (Religious(x) \lor Pious(x))]$$

Meaning: "There exists at least one person who is religious or pious."

iv) There is a barber who shaves all men in the town who do not shave themselves.

$$\exists x[Barber(x) \land \forall y((Man(y) \land \neg Shaves(y, y)) \rightarrow Shaves(x, y))]$$

Meaning: "There exists a barber x such that for every man y, if y does not shave himself, then x shaves y."

b) What is Resolution? Solve the following statement using Resolution Algorithm. Draw suitable resolution graph.

Resolution is an inference rule used in Artificial Intelligence and automated theorem proving to derive conclusions by refutation.

It works by:

1. Converting statements into clause form
2. Applying resolution rule
3. Deriving empty clause to prove statement

Resolution Rule

If: $P \lor Q$ and $\neg Q \lor R$

Then resolve on Q: $P \lor R$

i) Rajesh likes all kinds of food.

$$\forall x[Food(x) \rightarrow Likes(Rajesh, x)]$$

Clause form: $\neg Food(x) \lor Likes(Rajesh, x)$

ii) Apple and vegetables are food.

$$Food(Apple)$$
$$\forall x[Vegetable(x) \rightarrow Food(x)]$$

Clause form: $\neg Vegetable(x) \lor Food(x)$

iii) Anything anyone eats and is not killed is food.

$$\forall x \forall y[(Eats(x, y) \land \neg KilledBy(x, y)) \rightarrow Food(y)]$$

Clause form: $\neg Eats(x, y) \lor KilledBy(x, y) \lor Food(y)$

iv) Ajay eats peanuts and is still alive.

$$Eats(Ajay, Peanuts)$$
$$\neg KilledBy(Ajay, Peanuts)$$

Goal - Prove: $Likes(Rajesh, Peanuts)$

Negate goal: $\neg Likes(Rajesh, Peanuts)$

Resolution Steps

Step 1

From: $\neg Eats(x, y) \lor KilledBy(x, y) \lor Food(y)$

Substitute: $x = Ajay,\ y = Peanuts$

We get: $\neg Eats(Ajay, Peanuts) \lor KilledBy(Ajay, Peanuts) \lor Food(Peanuts)$

Step 2

Using: $Eats(Ajay, Peanuts)$

Resolve: $KilledBy(Ajay, Peanuts) \vee Food(Peanuts)$

Step 3

Using: $\neg KilledBy(Ajay, Peanuts)$

Resolve: $Food(Peanuts)$

Step 4

Using: $\neg Food(x) \vee Likes(Rajesh, x)$

Substitute: $x = Peanuts$

We get: $\neg Food(Peanuts) \vee Likes(Rajesh, Peanuts)$

Resolve with: $Food(Peanuts)$

Result: $Likes(Rajesh, Peanuts)$

Step 5 - Resolve with negated goal: $\neg Likes(Rajesh, Peanuts)$

Result: □ (Empty Clause)

Hence proved.

Resolution Graph -

Using resolution algorithm,

we proved: $Likes(Rajesh, Peanuts)$

Hence the statement is TRUE.

```
Eats(Ajay,Peanuts)
     \
      \
  ¬Eats(Ajay,Peanuts) ∨ KilledBy(Ajay,Peanuts) ∨ Food(Peanuts)
      /
     /
¬KilledBy(Ajay,Peanuts)
      ↓
  Food(Peanuts)
      ↓
¬Food(Peanuts) ∨ Likes(Rajesh,Peanuts)
      ↓
Likes(Rajesh,Peanuts)
      ↓
¬Likes(Rajesh,Peanuts)
      ↓
      □
```

Q4a) Write the logic

representation

for the following statements

i) "All birds fly"

$$\forall x[Bird(x) \rightarrow Fly(x)]$$

Meaning: "For every x, if x is a bird then x flies."

ii) "Every man respects his parents"

$$\forall x[Man(x) \rightarrow Respect(x, Parents(x))]$$

Meaning: "For every x, if x is a man then x respects his parents."

Alternative Representation

$$\forall x \forall y[(Man(x) \land Parent(y, x)) \rightarrow Respect(x, y)]$$

Meaning: "If y is parent of x and x is a man, then x respects y."

Q4**a) Detail the algorithm for deciding entailment in Propositional Logic.**

A sentence $\alpha$is entailed by knowledge base KBif $\alpha$is true in every model where KB is true.

It is represented as: $KB \vDash \alpha$ Meaning: "If KB is true, then αmust also be true."

Methods for Deciding Entailment - Truth Table Method, Inference Rules, Resolution Method

The most common algorithm is the **Truth Table Entailment Algorithm**.

Truth Table Entailment Algorithm

The algorithm checks whether query $\alpha$is true in all models where KB is true.

If yes: KB entails α

Otherwise: No entailment.

Steps of Algorithm

Step 1: Identify Symbols - Find all propositional symbols in: Knowledge Base (KB), Query (α)

Step 2: Generate Truth Table - Create all possible combinations of truth values.

For $n$symbols: 2^n rows are generated.

Step 3: Evaluate KB - Check rows where KB is TRUE.

Step 4: Evaluate Query - Check whether query is also TRUE in all rows where KB is TRUE.

Step 5: Decide Entailment

- If query is TRUE in every model of KB: $KB \vDash \alpha$

- Otherwise: $KB \nvDash \alpha$

Algorithm -

TT-ENTAILS(KB, α)

1. Generate all possible models
2. For each model:
   if KB is true in model:
      check α
      if α is false:
         return FALSE
3. Return TRUE

Example

Given: $KB = P \rightarrow Q$
P

Query: Q

Need to prove: $KB \vDash Q$

Truth Table -

P	Q	$P \rightarrow Q$	KB True?	Q
T	T	T	Yes	T
T	F	F	No	F
F	T	T	No	T
F	F	T	No	F

Only first row satisfies KB.

In that row: Q is TRUE.

Hence: $KB \vDash Q$

Therefore, entailment is proved.

Properties of Entailment Algorithm -

1. Sound - Produces only correct conclusions.
2. Complete - Finds all valid entailments.
3. Exhaustive - Checks all possible models.

Advantages - Simple and systematic, Guaranteed correct result, Useful for automated reasoning

Disadvantages - Time complexity is high, Requires 2^nrows, Inefficient for large knowledge bases

Applications - Expert systems, AI reasoning systems, Logic programming

Entailment in propositional logic determines whether a query logically follows from a knowledge base. The Truth Table algorithm systematically checks all possible models to verify logical correctness.